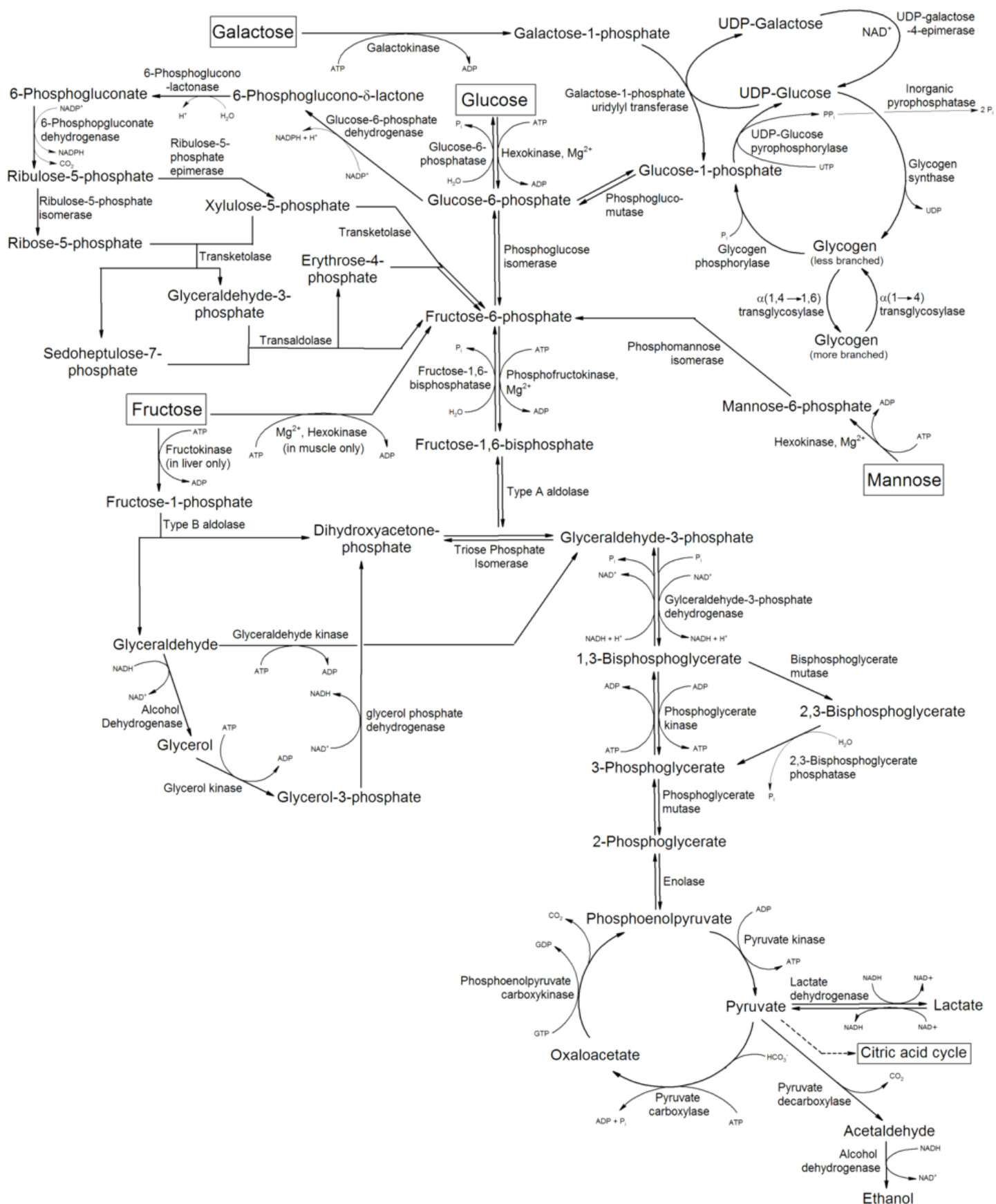




Glucose is the most abundant monosaccharide. Glucose is also the most widely used aldohexose in most living organisms. One possible explanation for this is that glucose has a lower tendency than other aldohexoses to react nonspecifically with the amine groups of proteins.^[29] This reaction—glycation—impairs or destroys the function of many proteins,^[29] e.g. in glycated hemoglobin. Glucose's low rate of glycation can be attributed to its having a more stable cyclic form compared to other aldohexoses, which means it spends less time than they do in its reactive open-chain form.^[29] The reason for glucose having the most stable cyclic form of all the aldohexoses is that its hydroxy groups (with the exception of the hydroxy group on the anomeric carbon of D-glucose) are in the equatorial position. Presumably, glucose